Kaustubh Hemant Sonawane

Bristol, UK | ksstubh.sonawane@gmail.com | +44 7388229743 | +91 8591441019 | www.github.com/Arishtanemi4 |
www.linkedin.com/in/kaustubhson4 | https://kaustubh-sonawane-portfolio.vercel.app

UK Visa Status: Student Visa. Eligible for Graduate Visa (2 Years Full Working Rights)

STATEMENT

AI Engineer with 3 years of industry experience building and deploying **LLM applications**, **RAG** services and **agentic tool-calling** workflows with **Azure OpenAI**, **LlamaIndex** and **pgvector**, designed end to end (React, FastAPI, AWS) for Compliance, Client Service and Investment teams in regulated financial services. Takes use cases from requirements through proof of concept to secure production release (department-level access control, LDAP **RBAC**, Compliance review, security-team **VAPT**, cost-aware model choice). Co-built a RAG-assisted scoring and ranking tool for an MSc dissertation with AstraZeneca and placed 2nd at the Encode hackathon with a spec-driven build. De facto team lead for up to 4 interns; twice recognised with Best Team Project.

EDUCATION

University of Bristol – MSc. Data Science (awaiting result) September 2025 – September 2026
NMIMS B.Tech Information Technology (Dual Degree) June 2017 – April 2022
NMIMS MBA Technology Management (Dual Degree) June 2017 – April 2022

WORK EXPERIENCE

HDFC Asset Management Company Ltd. July 2022 – July 2025

AI Solutions Engineer / Forward Deployed Engineer

GenAI, RAG & Agentic Systems

- Specified, designed and built **RAG** applications end to end (**ReactJS** frontend, **Python FastAPI** REST API microservices, **LlamaIndex** with **Azure OpenAI** GPT-3.5 & Llama-3) for Compliance, Client Service and Investment, using **Azure OpenAI embeddings** and a separate **pgvector** database per department so each team retrieved only from its own internal documents.
- Built **tool-calling** agents that call **Python** functions as tools to run **SQL**-backed fund comparisons and generate PDF reports on request; guided behaviour with **system prompts** versioned in **Git**, released to real users, changed the applications based on what they found, and had the Compliance department review each release.
- Weighed cost against quality when choosing models: **self-hosted Llama-3** on a **GPU EC2** instance for the largest department and the stronger **Azure OpenAI** outputs for decision makers, with the ability to switch models as needs changed.

Secure Production Delivery & Governance

- Turned prototypes into deployable tools: packaged the **FastAPI** microservices (uvicorn) as **Docker** containers on **AWS** (**EC2**, **ECR**) and automated deployment of the separate frontend and backend repositories (dev and prod) through a one-click **Jenkins** and **GitLab CI/CD** pipeline.
- Protected the internal applications with **RBAC** using a **Python** authentication function against the company **LDAP** server and **AWS Security Groups** limiting access to company private IPs; deployed an **Nginx** reverse proxy, one of the fixes that took the applications from repeatedly failing to passing the security team's **VAPT**.
- Coordinated with the security team on which ports to whitelist and which safeguards to maintain.

Stakeholder Delivery & Automation

- Worked directly with business stakeholders in Client Service, Compliance and Investment to turn requests into clear requirements, demonstrate working tools and turn feedback into releases; acted as de facto team lead for up to 4 interns in my final year; recognised with Best Team Project twice (2024 rating-email automation, 2025 department RAG chatbot).

- Built a **Python** extraction pipeline, scheduled with **EventBridge**, **Lambda** and **AWS SageMaker**, that uses **Azure OpenAI** to parse over **300** daily emails from credit rating agencies into structured data (first version used **AWS Textract**, replaced when a rating agency began sending emails as images), fuzzy-matches company names against the ones analysts track and sends a daily summary, saving **60** analyst hours a month.
- Designed a **Normalized SQL Schema** (**PostgreSQL**) and wrote **Python** ingestion scripts for the **Morningstar API** covering every fund in the Indian mutual fund market (2023–2025), integrated with legacy analyst data.
- Migrated financial analysts from Excel to **Power BI** dashboards, improving efficiency by **10** hours a week; analysed customer data using **Python Pandas** and **Market Basket Analysis** (**Apriori Algorithm**), handing the insights to the marketing team.

PROJECTS

- **University of Bristol MSc. Data Science Dissertation in partnership with AstraZeneca:** CellLineSelector (Team project, co-built. Data preprocessing pipeline, scoring algorithm and the RAG-based AI assistant that narrates results and answers methodology questions, in a tool built as FastAPI microservices with a React frontend that ranks cancer cell lines by target inclusion and exclusion genes and returns the evidence behind each pick; validated and refined with feedback from AstraZeneca scientists.) GitHub: https://github.com/Arishtanemi4/az-team25.git | Demo: http://celllineselector-frontend-408937187029.s3-website.eu-west-2.amazonaws.com/
- **Wingman:** AI dating-coach web app with 108 LLM personas across 12 archetypes, stored as data so tuning a persona needs no code change, and a vision model scoring uploaded photos (ReactJS, Python FastAPI, Kimi K2 and DeepSeek V4 Pro via NVIDIA NIM). LLM output treated as untrusted: validated with Pydantic, repaired and cached by input hash; rate limits shaped the design (12 calls instead of 108, serialised calls, results streamed to the UI). GitHub: https://github.com/Arishtanemi4/wingman.git

ACHIEVEMENTS

- **Encode Vibe Coding Hackathon:** Codeplain.ai track 2nd place winner. (Wingman.ai – An AI powered dating app simulation using codeplain's spec-driven framework) | GitHub: https://github.com/Arishtanemi4/wingman-codeplain.git
- **Project Hack 27**: Rolls Royce Derby 4th Place winner. (Human Centric Data Analytics tool to predict the competition winner based on human wellbeing and psychological temperament survey data.) | GitHub: https://github.com/Arishtanemi4/team-4b.git
- **University of Bristol Datathon sponsored by Lloyd's Bank:** 3rd Place winner. (Kaggle Credit card fraud detection problem.)

SKILLS

- **Tools:** Azure OpenAI, LlamaIndex, REST APIs, Git, Docker, Jenkins, GitLab (CI/CD), AWS (EC2 including a GPU instance, S3, ECR, Lambda, EventBridge, SageMaker), Nginx, Linux, LDAP, Claude Code (AI-assisted development), codeplain (spec-driven development), Power BI, VS Code.
- **Languages:** Python (pandas, scikit-learn, FastAPI, Pydantic, uvicorn), SQL, R (dplyr, tidyverse, ggplot2), JavaScript/TypeScript (NodeJS express, ReactJS), Shell
- **Databases:** PostgreSQL (pgvector semantic search, SQL functions and procedures), SQLite